

Movie Dataset Analysis Assignment

Objective

Use Pandas and Matplotlib to analyze the movie dataset and tell a story using graphs.

Task 1: Explore the Dataset

Load the dataset and display:

```
movies.head()

movies.info()

movies.describe()
```

Questions

1. How many movies are in the dataset?
 2. Which columns are numerical?
 3. Which columns are categorical?
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Task 2: Number of Movies Released Each Year

Use Pandas to determine the number of movies released each year.

Create a **line plot** to visualize the trend.

Questions

1. Which year has the most movies?
2. Does the number of movies increase over time?
3. What trend do you observe?

Task 3: Top 10 Movie Genres

Find the top 10 genres with the largest number of movies.

Create a **vertical bar chart**.

Questions

1. Which genre contains the most movies?
 2. Which genre contains the fewest movies among the top 10?
 3. Why is a bar chart suitable for this task?
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Task 4: Top 10 Directors

Find the top 10 directors with the largest number of movies.

Create a **horizontal bar chart**.

Questions

1. Which director produced the most movies?
 2. Why is a horizontal bar chart more suitable than a vertical bar chart for this visualization?
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Task 5: Distribution of Movie Ratings

Using the `score` column, create a **histogram**.

Questions

1. Which score range contains the largest number of movies?
 2. Are most movies highly rated?
 3. Are low-rated movies common?
 4. What does the histogram tell you about movie ratings?
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Task 6: Average Movie Score by Genre

Calculate the average score for each genre.

Sort the results from highest to lowest.

Create a **bar chart**.

Questions

1. Which genre has the highest average score?
 2. Which genre has the lowest average score?
 3. Are the most popular genres also the highest rated?
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Task 7: Top 10 Highest Grossing Movies

Find the top 10 movies with the highest gross revenue.

Create a **horizontal bar chart**.

Questions

1. Which movie earned the highest revenue?
 2. Is there a large difference between the highest-grossing movie and the others?
 3. What observations can you make from the graph?
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Final Storytelling Question

Imagine you are a data analyst working for a movie production company.

Write a short report (5–8 sentences) describing your findings.

Your report should answer the following questions:

1. Which genre is the most common?
2. Which genre has the highest average rating?
3. Are most movies highly rated?
4. Has movie production increased over time?
5. Which movie earned the highest revenue?

6. What recommendations would you give to a movie studio based on your analysis?